

Exploratory Data Analysis

```
library(tidyverse)
```

```
— Attaching core tidyverse packages ————— tidyverse 2.0.0
—
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.1    ✓ stringr    1.5.2
✓ ggplot2     4.0.0    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr       1.1.0
— Conflicts ————— tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(readr)
library(sf)
```

Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.0; sf_use_s2() is TRUE

```
library(tigris)
```

To enable caching of data, set `options(tigris\_use\_cache = TRUE)`
in your R script or .Rprofile.

```
library(dplyr)
library(ggplot2)
library(stats)
precinct_2024_general_election <- read.csv("data/g24_sov_by_g24_svprec.csv")

simpler_precinct <- precinct_2024_general_election |>
  select("COUNTY", "TOTVOTE", "CDDIST", "ASSDEM01", "ASSDEM02", "ASSREP01",
         "ASSREP02", "CNGDEM01", "CNGDEM02", "CNGREP01", "CNGREP02",
         "PRSDEM01", "PRSREP01", "SENDEM01", "SENDEM02",
         "SENREP01", "SENREP02", "USSDEM01", "USSREP01") |>
  filter(
```

```

!if_any(
  .cols = -COUNTY,
  .fns = ~ . == "***"
)) |>
  mutate(
    across(
      .cols = -COUNTY,
      .fns = ~ as.integer(as.character(.))
    )
  )

```

Question 1

How many votes for the presidential democrat vs republican candidate had a less than 1 percent difference in vote count between them?

Answer 1

```

closest_race_analysis <- simpler_precinct |>

mutate(
  Contest_Total = PRSDEM01 + PRSREP01,
  Raw_Difference = abs(PRSDEM01 - PRSREP01),
  Pct_Margin = (Raw_Difference / Contest_Total) * 100
)

closest_precinct <- closest_race_analysis |>
  arrange(Pct_Margin)

```

```

count_under_1_percent <- closest_race_analysis |>
  filter(Pct_Margin < 1) |>
  count()
count_under_1_percent

```

```

n
1 782

```

782

Question 2

What is the range of total sizes of the precincts?

Answer 2

```

precinct_size_summary <- simpler_precinct |>
  summarise(

```

```

Min_Total_Votes = min(TOTVOTE),
Max_Total_Votes = max(TOTVOTE),
Mean_Total_Votes = mean(TOTVOTE),
Median_Total_Votes = median(TOTVOTE),
Range_of_Votes = Max_Total_Votes - Min_Total_Votes
)
print(precinct_size_summary)

```

```

Min_Total_Votes Max_Total_Votes Mean_Total_Votes Median_Total_Votes
1              0          3793980          2424.274             121
Range_of_Votes
1          3793980

```

The range goes from a vote count of 0 to 3793980, with a mean of 2424.274 votes and median of 121 votes.

Question 3

How large were the largest counties and how small were the smallest?

Answer 3

```

county_precinct_counts <- simpler_precinct |>
  group_by(COUNTY) |>
  count(name = "Precinct_Count") |>
  arrange(desc(Precinct_Count))
print(county_precinct_counts)

```

```

# A tibble: 57 × 2
# Groups:   COUNTY [57]
  COUNTY Precinct_Count
  <int>      <int>
1     19         5937
2     37         5117
3     36         5074
4     30         4558
5     56         2459
6     33         2433
7      7         2037
8     43         1498
9     15         1480
10    34         1349
# i 47 more rows

```

Largest county is labeled county 19 with 5937 precincts.

```

county_precinct_counts <- simpler_precinct |>
  group_by(COUNTY) |>
  count(name = "Precinct_Count") |>
  arrange((Precinct_Count))
print(county_precinct_counts)

```

```

# A tibble: 57 × 2
# Groups:   COUNTY [57]
  COUNTY Precinct_Count
  <int>         <int>
1     2             19
2     6             33
3    46             33
4    53             34
5    26             36
6    32             43
7    14             44
8    22             48
9    25             48
10     8             50
# i 47 more rows

```

Smallest is county 2 with 19 precincts.